

30

Technology Focus
Robots That 'Grow Up'



12

eStyle Buyers' Guide
Smartphone Cameras:
It's All About What Clicks



22 Technology Focus

The Robots' Guide To World Domination

36 Innovation

A Combo Connectivity Chipset for IoT Applications

44 Innovation

Part 1 of 2: An Introduction to Wireless Sensor Networks

52 Defence Electronics

Part 1 of 3: The Electronics Behind Big Guns

58 Telecom Technology

FTTH and Passive Optical Networks

62 Embedded Design

What's New in Software Design Tools for Embedded Systems

68 Viewpoint

Incentives Restricting Solar Business in India

72 Renewable Energy

Solar-Powered Irrigation Systems

78 Renewable Energy

Efficiently Harnessing Solar Power With Tracking System

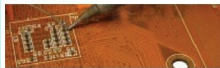
82 Renewable Energy

Energy Education and Research Initiatives in India

84 EFY Plus DVD

This Month's DVD Contents

Hands-On



DIY: Project

- XBee-Controlled Aircraft
- Celestial Weight Calculator

DIY: Circuit

- Desktop LED Emergency Light
- Simple Interface for Digital Sound Synthesis
- Tester for 555 Timer and 741 Op-amp ICs
- Lithium-Ion Battery Charger

DIY: Tips & Technique

- Surveillance Using Raspberry Pi Camera
- Web Server Setup for WordPress on Raspberry Pi

DIY: Software

- Face Counter Using MATLAB
- Getting Started With Kinect and OpenCV
- Running Linux on Windows

85 Test & Measurement

Today's Changing Instrumentation World

90 Career

Instrumentation Offers Most Jobs in Marketing and Software

124 Career

Embedded World Offers Plethora of Jobs But With Low Starting Pay

129 Industry Focus

Solar Industry Expected to Grow Fast

144 Interview

"Each manufacturer has its own strategy on what to test and where"—Lauri Antila, vice president-marketing and business development, JOT Automation

Regulars

8 First Look	139 New Products	157 Electronics Mart Ads
16 Technology News	141 Letters	162 Product Categories Index
123 Useful Websites	142 Q&As	
136 Industry News	145 Business Pages Ads	163 Advertisers' Index

EFY Plus DVD

NI Circuit Design Suite For All Your Circuit Designing Needs
I KTeclab: An Integrated Development Environment
N DesignSpark: For All Your PCB Designing Needs
VI

NEXT ISSUE

• Open Source Electronics

• Educational & Training Products

• Development Boards (Microcontroller based)

EDITOR	: Ramesh Chopra
EDITORIAL SECRETARY	: Editorial Secretary
CORRESPONDENCE	: Phone: 011-26810601; E-mail: editor@efy.in (Technical queries: eflyablog@efy.in)
SUBSCRIPTIONS & MISSING ISSUES	: Phone: 011-26810601 or 02 or 03 E-mail: support@efy.in
BACK ISSUES, BOOKS, CDs, PCBs etc.	: Kite's Spares, New Delhi Phone: 011-26371661, 26371662 E-mail: info@kitespares.com
EXCLUSIVE NEWSSTAND DISTRIBUTOR	: BH Books & Magazine Distributors Ltd, Mumbai Phone: 022-40497401, 40497402, 40497474, 40497413; Fax: 40497434 E-mail: circulations@bhbworld.com
ADVERTISEMENTS	: Ph: 011-26810601 or 02 or 03 NEW DELHI (HEAD OFFICE) E-mail: eflyadvertising@efy.in
MUMBAI	: Ph: 022-24950047, 24928520 E-mail: eflymumbai@efy.in
BENGALURU	: Ph: 080-25260394, 25260023 E-mail: eflyb@efy.in
CHENNAI	: Ph: 044-42994363 E-mail: eflychennai@efy.in
HYDERABAD	: Ph: 040-67172633 E-mail: eflyhyderabad@efy.in
KOLKATA	: Ph: 033-22294788 E-mail: eflykolkata@efy.in
PUNE	: Ph: 020-40147882 E-mail: eflypune@efy.in
GUJARAT	: Ph: 079-61344948 E-mail: eflyahmedabad@efy.in
CHINA	: Power Pioneer Group Inc. Ph: (86 755) 83729797, (86) 13923802595 E-mail: powerpioneer@efy.in
JAPAN	: Tandem Inc., Ph: 81-3-3541-4166 E-mail: tandem@efy.in
SINGAPORE	: Publicitas Singapore Pte Ltd Ph: +65-68362272 E-mail: publicitas@efy.in
TAIWAN	: J.K. Media, Ph: 886-2-87726780 ext. 10 E-mail: jkmedia@efy.in
UNITED STATES	: E & Tech Media Ph: +1 860 536 6677 E-mail: veneniquelamange@gmail.com

Printed, published and owned by Ramesh Chopra. Printed at Nucleus Photolithographers, 8-38, Okhla Industrial Area, Phase-1, New Delhi 110020. Copyright 2014. All rights reserved throughout the world. Reproduction of any material from this magazine in any manner without the written permission of the publisher is prohibited. Although every effort is made to ensure accuracy, no responsibility whatsoever is taken for any loss due to publishing errors. Articles that cannot be used are returned to the authors accompanied by a self-addressed and sufficiently stamped envelope. But no responsibility is taken for any loss or delay in returning the material. EFL will not be responsible for any wrong claims made by an advertiser. Disputes, if any, will be settled in a New Delhi court only.